



Security Assessment

Draft Report

ROYCO

Royco Dawn

April 2026

Prepared for Royco

Table of Contents

Project Summary	3
Project Scope	3
Project Overview	3
Protocol Overview	4
Assessment Methodology	5
Threat and Security Overview	6
Findings Summary	7
Severity Matrix	7
Detailed Findings	8
Medium Severity Issues	10
M-01: JT deposits at jtEffectiveNAV = 0 can permanently brick JT recapitalizations	10
M-02: Soulbound tranche kernels are incompatible with RoycoEntryPoint	12
Low Severity Issues	13
L-01: Admin-controlled tranche share seizure is blocked when the tranche is paused	13
L-02: Race condition between RoycoEntryPoint cancel and execute functions	14
L-03: Unsafe cast in Units.toInt256	15
L-04: Synchronous protocol fee collection allows DoS through fee recipient external blacklisting	16
L-05: Self-liquidation bonus can increase utilization be incorrect for beta > 1	17
L-06: stSelfLiquidationBonusWAD should be validated to not exceed WAD	18
L-07: Missing TRANCHE_TYPE validation in RoycoFactory	19
L-08: Delayed ST redemptions don't forfeit appreciation to self-liquidation bonus	20
L-09: Undistributed yield of old YDM can cause incorrect yield distribution after a new YDM is configured	21
L-10: Blacklisted users can exit escrowed tranche shares through `executeRedemption()`	22
Informational Severity Issues	23
I-01: Market's fixedTermEndTimestamp is never reset	23
I-02: RoycoEntryPoint could allow executions across multiple users	24
I-03: Changes to RoycoEntryPoint's yieldRecipient are retroactive	25
I-04: RoycoEntryPoint can force many dust-sized executions when close to tranche max	26
I-05: Tranche deposits can succeed with zero shares	27
I-06: Dust guard not re-checked after `jtNetGain` is reduced by coverage	28
Disclaimer	29
About Certora	29

Project Summary

Project Scope

Project Name	Repository Link	Commit Hash	Platform
Royco Dawn	https://github.com/roycoprotocol/royco-dawn/tree/feat/entry-point	85bf7dc	Solidity

Project Overview

This document describes the security review of **Royco Dawn**. The work was undertaken from **March 30th, 2026** to **April 16th, 2026**.

The following contract list is included in our scope:

```
src/auth/RoycoAuth.sol
src/base/RoycoBase.sol
src/libraries/Constants.sol
src/libraries/Types.sol
src/libraries/UtilsLib.sol
src/libraries/Units.sol
src/periphery/RoycoMarketSyncer.sol
src/periphery/RoycoEntryPoint.sol
src/accountant/RoycoAccountant.sol
src/factory/RoycoFactory.sol
src/factory/RolesConfiguration.sol
src/tranches/RoycoSeniorTranche.sol
src/tranches/RoycoJuniorTranche.sol
src/tranches/base/RoycoVaultTranche.sol
src/ydm/AdaptiveCurveYDM_V2.sol
src/kernels/base/RoycoKernel.sol
src/kernels/base/quoter/IdenticalERC4626SharesToAdminOracleQuoter.sol
src/kernels/base/quoter/IdenticalMakinaSharesToAdminOracleQuoter.sol
```

```
src/kernels/base/quoter/IdenticalERC4626SharesToChainlinkOracleQuoter.sol
src/kernels/base/quoter/IdenticalAssetsChainlinkToAdminOracleQuoter.sol
src/kernels/base/quoter/base/IdenticalAssetsOracleQuoter.sol
src/kernels/base/quoter/base/IdenticalAssetsChainlinkOracleQuoter.sol
src/kernels/base/quoter/base/IdenticalERC4626SharesOracleQuoter.sol
src/kernels/base/quoter/base/IdenticalAssetsAdminOracleQuoter.sol
src/kernels/Identical_Makina_ST_JT_MachineToAdminOracle_Kernel.sol
src/kernels/sUSDai_ST_JT_RedemptionSharePriceToAdminOracle_Kernel.sol
src/kernels/MaplePoolV2_ST_JT_ExitSharePriceToChainlinkOracle_Kernel.sol
src/kernels/Identical_ERC20_ST_JT_ChainlinkToAdminOracle_SoulBoundTrancheS
hares_Kernel.sol
src/kernels/ReUSD_ST_JT_ICLOracle_Kernel.sol
src/kernels/Identical_ERC4626_ST_JT_SharePriceToChainlinkOracle_Kernel.sol
src/kernels/Identical_AA_IdleCD0_ST_JT_VirtualPriceOracle_Kernel.sol
src/kernels/sUSDat_ST_JT_SharePriceToChainlinkOracle_Kernel.sol
src/kernels/Identical_ERC4626_ST_JT_SharePriceToAdminOracle_Kernel.sol
src/kernels/Identical_ERC20_ST_JT_ChainlinkToAdminOracle_Kernel.sol
```

The team performed a manual audit of all the files in scope. During the manual audit, the Certora team discovered bugs in the code, as listed on the following page.

Protocol Overview

Royco Dawn is a tranche-based structured product system where an ERC-20 vault, called Junior Tranche (JT), provides another vault (Senior Tranche, ST) with risk coverage against market fluctuation in exchange for a share of the yield accrued by the protected (ST) vault.

The protocol consists of a factory/access manager that is common to all markets, and multiple markets composed of senior and junior tranche ERC20 vaults, a kernel, an accountant, and YDM modules, with an async EntryPoint on top.

Once users deposit assets into the ST or JT, these are transferred to the kernel; the kernel then syncs the accountant, which marks raw net asset value (NAV) to market, tracks effective NAV and impermanent-loss style liabilities, enforces coverage, and allocates ST gains to JT through the yield distribution model (YDM).

Assessment Methodology

Our assessment approach combines design level analysis with a deep review of the implementation to ensure that a protocol is secure, economically sound, and behaves as intended under realistic conditions.

At the design level, we evaluate the architecture, the economic assumptions behind the protocol, and the safety properties that should hold independently of a specific chain or environment. This process includes reviewing internal and cross protocol interactions, state transition flows, trust boundaries, and any mechanism that could be exploited to extract value, deny service, or alter core system behavior. At this stage, a focused threat modelling exercise helps identify key attack surfaces and adversarial capabilities relevant to the system. Design level issues often relate to incentive structures, governance implications, or systemic behavior that emerges under adversarial conditions.

Implementation analysis focuses on the concrete behavior of the code within the execution model of the target chain. This involves reviewing the correctness of logic, access control, state handling, arithmetic behavior, and the nuanced behaviors of the chain environment. Familiar classes of vulnerabilities such as reentrancy conditions, faulty permission checks, precision issues, or unsafe assumptions often surface at this layer. These findings require context aware reasoning that takes into account both the code and the architectural intent.

To support this analysis, the codebase is examined through repeated manual passes and supplemented by automated tools when appropriate. High-risk logic areas receive deeper scrutiny, invariants are validated against both design intent and actual implementation, and potential vulnerability leads are thoroughly investigated. Automated techniques such as static analysis, fuzzing, or symbolic execution may be used to complement manual review and provide additional insight.

Collaboration with the development team plays an important role throughout the audit. This helps confirm expected behaviors, clarify design assumptions, and ensure an accurate understanding of the protocol's intended operation. All findings are documented with clear reasoning, reproducible examples, and actionable recommendations. A follow up review is conducted to validate the applied fixes and verify that no regressions or secondary issues have been introduced.

Threat and Security Overview

The main trust boundaries are the factory-backed role system, oracle/quoter admins, kernel/accountant admins, sync operators, and external integrations such as Chainlink and ERC4626-style vaults. The core invariants are NAV conservation, correct coverage/utilization enforcement, coherent fixed-term/perpetual state transitions, consistent delayed request valuation, and stable tranche share semantics under rounding.

The audit work focused most heavily on [RoycoAccountant](#), [RoycoKernel](#), [RoycoVaultTranche](#), [RoycoEntryPoint](#), the [AdaptiveCurveYDM_V2](#) implementation, and the factory/roles setup, with cross-checks against deployment scripts and market configs. Special attention was given to the transition from permissioned participation by selected actors in the markets to a permissionless setup where anyone can participate.

Existing coverage in the repository is broad: accountant unit/comprehensive tests, YDM tests, abstract kernel fuzz suites, protocol-specific integration tests, entrypoint tests, and upgradeability/roles/pausability tests.

Findings Summary

The table below summarizes the findings of the review, including type and severity details.

Severity	Discovered	Confirmed	Fixed
Critical	-	-	-
High	-	-	-
Medium	2	-	-
Low	10	-	-
Informational	6	-	-
Total	18	-	-

Severity Matrix

Impact	High	Medium	High	Critical
	Medium	Low	Medium	High
	Low	Low	Low	Medium
		Low	Medium	High
Likelihood				

Detailed Findings

ID	Title	Severity	Status
M-01	JT deposits at jtEffectiveNAV = 0 can permanently brick JT recapitalizations	Medium	
M-02	Soulbound tranche kernels are incompatible with RoycoEntryPoint	Medium	
L-01	Admin-controlled tranche share seizure is blocked when the tranche is paused	Low	
L-02	Race condition between RoycoEntryPoint cancel and execute functions	Low	
L-03	Unsafe cast in Units.toInt256	Low	
L-04	Synchronous protocol fee collection allows DoS through fee recipient external blacklisting	Low	
L-05	Self-liquidation bonus can increase utilization be incorrect for beta > 1	Low	
L-06	stSelfLiquidationBonusWAD should be validated to not exceed WAD	Low	
L-07	Missing TRANCHE_TYPE validation in RoycoFactory	Low	

L-08	Delayed ST redemptions don't forfeit appreciation to self-liquidation bonus	Low	
L-09	Undistributed yield of old YDM can cause incorrect yield distribution after a new YDM is configured	Low	
L-10	Blacklisted users can exit escrowed tranche shares through <code>executeRedemption()</code>	Low	
I-01	Market's <code>fixedTermEndTimestamp</code> is never reset	Informational	
I-02	<code>RoycoEntryPoint</code> could allow executions across multiple users	Informational	
I-03	Changes to <code>RoycoEntryPoint</code> 's <code>yieldRecipient</code> are retroactive	Informational	
I-04	<code>RoycoEntryPoint</code> can force many dust-sized executions when close to tranche max	Informational	
I-05	Tranche deposits can succeed with zero shares	Informational	
I-06	Dust guard not re-checked after <code>jtNetGain</code> is reduced by coverage	Informational	

Medium Severity Issues

M-01: JT deposits at $jtEffectiveNAV = 0$ can permanently brick JT recapitalizations

Severity: Medium	Impact: Medium	Likelihood: Medium
Files: RoycoVaultTranche.sol	Status:	

Description:

When users deposit into vault tranches, the shares they get in return are calculated proportionally to the pre-existing shares (`totalSupply`) and net value of the tranche assets (`effectiveNAVToMintAt`).

For Junior tranches, this `effectiveNAVToMintAt` can become zero when the tranche has allocated all its value in support of its Senior tranche, while `totalSupply` remains unchanged from previous deposits.

This means that every time a recapitalization (a deposit with `jtEffectiveNAV == 0 && totalSupply() > 0`) happens, the JT share conversion rate gets diluted by 18+ orders of magnitude.

If over the lifetime of a market, a recapitalization event like this happens a few times (typically 3 or 4), the share price can become diluted to the point of overflowing even for reasonably-sized deposits.

Recommendations:

Consider resetting JT share ownerships under extreme dilution (the stRSR token is a known example of implementing the concept of “eras”).

Customer Response:

Awaiting a response

Fix Review:

Awaiting a response

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M-02: Soulbound tranche kernels are incompatible with RoycoEntryPoint

Severity: Medium	Impact: Medium	Likelihood: Medium
Files: Identical_ERC20_ST_JT_ChainlinkToAdminOracle_SoulBoundTrancheShares_Kernel.sol	Status:	

Description:

The **SoulBound** kernel uses the **_preTrancheBalanceUpdate** hook to prevent transfer of tranche shares other than fee attributions.

This limitation is, however, too strict as it does not account for the case of markets requiring an intermediate **RoycoEntryPoint** transfer for entries and exits.

Recommendations:

Consider adding an optional **RoycoEntryPoint** extra exception to the **_preTrancheBalanceUpdate** checks.

Customer Response:

Awaiting a response

Fix Review:

Awaiting a response

Low Severity Issues

L-01: Admin-controlled tranche share seizure is blocked when the tranche is paused

Severity: Low	Impact: Medium	Likelihood: Low
Files: RoycoVaultTranche.sol	Status:	

Description:

The `seizeShares` and `seizeAndRedeemShares` functions of the Royco tranches internally call `ERC20PausableUpgradeable._update` that is decorated with the `whenNotPaused` modifier.

This means that when a tranche is paused, these functions are not callable despite requiring a high privilege.

While it is not excluded that an admin can work around this limitation by multicalling `unpause + seize + pause` in a single transaction to perform the operation, it may not always be the case, i.e., if operating through an EOA.

Recommendations:

Consider having `ERC20Upgradeable._update` instead of `super._update` in `seizeShares` and `seizeAndRedeemShares`.

Customer Response:

Awaiting a response

Fix Review:

Awaiting a response

L-02: Race condition between RoycoEntryPoint cancel and execute functions

Severity: Low	Impact: Low	Likelihood: Low
Files: RoycoEntryPoint.sol	Status:	

Recommendations:

The **RoycoEntryPoint** contract allows users to queue deposit and redemption requests for an executor to pick up at a later point in time.

Executors, however, batch requests together, and at the same time, users can cancel some of their requests at any time.

It is therefore possible that a batch fails because one of the requests has been cancelled shortly before the batch is executed.

Customer Response:

Consider having the deposit and redemption execution batches skip without failing when one of the batched requests has been cancelled or filled already.

Fix Review:

Awaiting a response

L-03: Unsafe cast in Units.toInt256

Severity: Low	Impact: Low	Likelihood: Low
Files: Units.sol	Status:	

Description:

The `Units.toInt256` function performs a cast from `NAV_UNIT (uint256)` to `int256`. This cast can be lossy if the converted `_unit` exceeds `type(int256).max`.

While we don't see an extreme scenario where this loss can happen, we recommend defending in depth against overflows.

Recommendations:

Consider explicitly checking for `_units` exceeding `type(int256).max`.

Customer Response:

Awaiting a response

Fix Review:

Awaiting a response

L-04: Synchronous protocol fee collection allows DoS through fee recipient external blacklisting

Severity: Low	Impact: Medium	Likelihood: Low
Files: RoycoKernel.sol	Status:	

Description:

In the `_preOpSyncTrancheAccounting` hook, called in all `deposit` and `redeem` operations, fees are collected and synchronously minted to the `protocolFeeRecipient` address.

This minting will, in turn, validate the operation through the kernel's `_preTrancheBalanceUpdate` that checks for `protocolFeeRecipient` to be an acceptable recipient for the underlying asset.

This means that if `protocolFeeRecipient` is at any point blacklisted or in any other way disallowed to receive amounts of underlying asset, the whole market is frozen as the fee minting fails, bubbling up the revert into `_preOpSyncTrancheAccounting` and the liquidity operation that triggered it.

Recommendations:

Consider minting fees synchronously to the Kernel instead, and allowing `protocolFeeRecipient` to pull them asynchronously at a later point. This would reduce the system-wide exposure to blacklisting to what is strictly necessary, which is the tranches themselves.

Customer Response:

Awaiting a response

Fix Review:

Awaiting a response

L-05: Self-liquidation bonus can increase utilization be incorrect for $\beta > 1$

Severity: Low	Impact: Medium	Likelihood: Low
Files: RoycoKernel.sol	Status:	

Description:

The `RoycoAccountant` configuration parameters allow for `betaWAD` to be configured above 100%, and this conceptually makes sense in case JT and ST underlying are correlated through a leverage factor.

When `betaWAD` is configured above 100%, the math in `RoycoKernel._computeMaxUtilizationNeutralBonus` deviates from its theoretical formulation (`RoycoKernel.sol:L920`) because of the use of a defensive `saturatingSub` at L922, that is a non-linear component that is not present in the theoretical formula.

Because `saturatingSub` returns a higher value for `WAD - beta` than the theoretical math would, and its result is positively contributing to the returned max bonus, the resulting max bonus can be overestimated, which in turn can allow for the self-liquidation bonus to exceed the utilization-neutral threshold.

Recommendations:

Consider either linearizing the formula, i.e. with a $a > b ? a - b : -(b - a)$ term, or capping beta to 100%, which is a safe choice for the current scope where all kernels are same-asset.

Customer Response:

Awaiting a response

Fix Review:

Awaiting a response

L-06: stSelfLiquidationBonusWAD should be validated to not exceed WAD

Severity: Low	Impact: Low	Likelihood: Low
Files: RoycoKernel.sol	Status:	

Description:

The `stSelfLiquidationBonusWAD` percentage parameter is set in `RoycoKernel` through the `__RoycoKernel_init` initializer and the `setSeniorTrancheSelfLiquidationBonus` setter. In both cases, its value is not validated to not exceed `WAD`.

Recommendations:

Consider adding a validation for the to-be-configured value to not exceed `WAD`.

Customer Response:

Awaiting a response

Fix Review:

Awaiting a response

L-07: Missing TRANCHE_TYPE validation in RoycoFactory

Severity: Low	Impact: Medium	Likelihood: Low
Files: RoycoFactory.sol	Status:	

Description:

After deploying a market, **RoycoFactory** validates the creation and initialization of the various contracts against its expectations through its `_validateDeployment` function.

This function, however, does not validate that for the Junior and Senior tranches, a correct implementation (Junior or Senior, respectively) is being used.

Recommendations:

Consider adding to the `_validateDeployment` function checks for the values returned by `_roycoMarket.seniorTranche.TRANCHE_TYPE()` and `_roycoMarket.juniorTranche.TRANCHE_TYPE()`.

Customer Response:

Awaiting a response

Fix Review:

Awaiting a response

L-08: Delayed ST redemptions don't forfeit appreciation to self-liquidation bonus

Severity: Low	Impact: Medium	Likelihood: Low
Files: RoycoEntryPoint.sol	Status:	

Description:

The `RoycoEntryPoint` is designed to optionally capture the yield accrued by tranche shares between the moment when their redemption is requested and when it's executed.

It does so by snapshotting the `convertToAssets` output and comparing it with the same output at the time of execution, to seize any appreciation that could have occurred.

This calculation, however, does not include the self-liquidation bonus that can be applied to ST shares, allowing part of the appreciation to be sheltered from seizure.

Recommendations:

Consider using `previewRedeem` instead of `convertToAssets` to measure share appreciation during the escrowed period.

Customer Response:

Awaiting a response

Fix Review:

Awaiting a response

L-09: Undistributed yield of old YDM can cause incorrect yield distribution after a new YDM is configured

Severity: Low	Impact: Medium	Likelihood: Low
Files: RoycoAccountant.sol	Status:	

Description:

When a market is updated with a new YDM (`setYDM` function), it first syncs the market to make sure the new YDM starts working in a clean state. There can, however, be a case when a stale state from the old YDM contaminates the math for the new YDM, that is when at the time of the `setYDM` call, interest was accrued but not distributed because below the dust threshold.

This leaves the market's `twJTYieldShareAccruedWAD` and `lastDistributionTimestamp` state variables in a state that is not clean (i.e. other than `0` and `block.timestamp` respectively), exposing new yield accrued to a potentially significant difference in attribution.

Recommendations:

Consider forcing `twJTYieldShareAccruedWAD` and `lastDistributionTimestamp` to `0` and `block.timestamp` respectively when `setYDM` is called, so the skipped dust distribution does not impact future yield. This action would sacrifice dust yield to avoid a potentially significant impact on future yield.

Customer Response:

Awaiting a response

Fix Review:

Awaiting a response

L-10: Blacklisted users can exit escrowed tranche shares through `executeRedemption()`

Severity: Low	Impact: Medium	Likelihood: Low
Files: RoycoEntryPoint.sol	Status:	

Description:

`requestRedemption()` on `RoycoEntryPoint` transfers the user's tranche shares into the contract for the duration of `redemptionDelaySeconds`. At execution time, `executeRedemption()` calls `IRoycoVaultTranche(tranche).redeem(userSharesRedeemed, _receiver, address(this))` via `_redeemWithYieldRouting()`, so the kernel's `preTrancheBalanceUpdateHook()` is invoked with `_caller = EntryPoint`, `_from = EntryPoint` and `_to = address(0)`. The original requester (`_user`) is never part of the call graph, and the underlying assets paid out by the kernel's `_withdrawAssets()` to `_receiver` are plain ERC20 transfers that do not invoke the hook at all.

As a result, if an admin blacklists the requester after they have queued a redemption, the existing request remains fully executable. The blacklisted user (or any keeper on their behalf, given the delay has elapsed) can still call `executeRedemption()` and withdraw the underlying ST/JT assets to `_receiver`, defeating the kernel's blacklist as a compliance control for any account with a pending redemption.

Recommendations:

In `executeRedemption()`, explicitly check that both `_user` (the original requester) and `request.baseRequest.receiver` are not blacklisted on the tranche's kernel before redeeming the escrowed shares.

Customer Response:

Awaiting a response

Fix Review:

Awaiting a response

Informational Severity Issues

I-01: Market's `fixedTermEndTimestamp` is never reset

Files:

RoycoAccountant.sol

Description:

When a market enters the `FIXED_TERM` state, `fixedTermEndTimestamp` is set to the point in the future after which it will transition back `PERPETUAL`. However, when this transition happens `fixedTermEndTimestamp` is re-computed to zero by `RoycoAccountant` but never persisted.

Recommendations:

Consider persisting `fixedTermEndTimestamp` also in its non-zero to zero transitions.

Customer Response:

Awaiting a response

Fix Review:

Awaiting a response

I-02: RoycoEntryPoint could allow executions across multiple users

Files:

RoycoEntryPoint.sol

Description:

The `executeDeposits` and `executeRedemptions` functions of `RoycoEntryPoint` accept multiple operations of a single user for batch execution.

If these functions accepted users in an array instead of a single `address`, batching could be further optimized to span several accounts without requiring multiple calls.

Recommendations:

Consider modifying the `executeDeposits` and `executeRedemptions` functions to accept `users` in an array.

Customer Response:

Awaiting a response

Fix Review:

Awaiting a response

I-03: Changes to RoycoEntryPoint's yieldRecipient are retroactive

Files:

RoycoEntryPoint.sol

Description:

The `RoycoEntryPoint` contract allows admins to change the yield recipient (`modifyTrancheConfigs`) of a given tranche. This change, however, has a retroactive impact on requests that are posted but not yet executed at the time of the config modification.

Recommendations:

Consider snapshotting the `yieldRecipient` config with the user requests.

Customer Response:

Awaiting a response

Fix Review:

Awaiting a response

I-04: RoycoEntryPoint can force many dust-sized executions when close to tranche max

Files:

RoycoEntryPoint.sol

Description:

The deposit and redemption logic in `RoycoEntryPoint` caps executions at the tranche's declared `maxDeposit` and `maxRedeem`.

When this cap is applied, and in case a fee is defined for the operation, the amount effectively deposited or redeemed can be further reduced by the shares kept as fee.

This has two consequences – the most immediate one is that an unnecessarily smaller share of the operation is executed. In case of batched requests though, the portion of `maxDeposit` (or `maxRedeem`) that was unused because of fees will most likely be allocated to the next transaction, which will deposit that amount minus fees. Because fees are orders of magnitude smaller than requests, a batch close to the Tranch maximum will see its requests executed with a fat-tail of smaller and smaller amounts, quickly reaching the point where they become irrelevant.

Recommendations:

Consider adjusting the logic that caps to `maxDeposit` and `maxRedeem` to apply the cap to the amount after, not before, the fee, so the cap can be consumed by the earliest elements of the batch, in full when applicable.

Customer Response:

Awaiting a response

Fix Review:

Awaiting a response

I-05: Tranche deposits can succeed with zero shares

Files:

RoycoVaultTranche.sol

Description:

The `RoycoVaultTranche.deposit` function implements a safeguard for preventing users from depositing `ZERO_NAV_UNITS` of value allocated.

Because they are not specifying a `minSharesOut` requirement, it would be desirable that `shares` are validated to be non-zero, because a non-zero allocated value can still translate to zero `shares` after the `_convertToShares` conversion is applied.

Recommendations:

Consider replacing the `valueAllocated != ZERO_NAV_UNITS` requirement with a slightly stricter `shares != 0`.

Customer Response:

Awaiting a response

Fix Review:

Awaiting a response

I-06: Dust guard not re-checked after `jtNetGain` is reduced by coverage

Files:

RoycoAccountant.sol

Description:

`RoycoAccountant._previewSyncTrancheAccounting()` checks `jtNetGain > jtNAVDustTolerance` before computing JT protocol fees, but when later ST-loss coverage reduces `jtNetGain` via `saturatingSub`, the code recomputes `jtProtocolFeeAccrued` without reapplying the dust guard. As a result, a gain that was initially above dust can be pushed back into the dust range and still flow through fee math, violating the intended “no fees on dust” logic. In practice the impact is negligible because floor rounding usually collapses the fee to zero, but the accounting logic is internally inconsistent.

Recommendations:

Consider re-applying the dust guard when `jtProtocolFeeAccrued` is calculated.

Customer Response:

Awaiting a response

Fix Review:

Awaiting a response

Disclaimer

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